

CLINOMA PLATFORM

# FIRST PAPER CAMP

## Questions & Exercises Booklet

Day 1: Pediatrics

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## CHAPTER 01

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# Growth and Development (Selected Topics)

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## 1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four clinical uses of Percentile Curves in growth charts.

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Q2: Enumerate four key development warning signs in children.

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## 2. Short Answered Questions (Definitions, Mention, List)

Q3: Define the terms "Growth" and "Development" according to your curriculum.

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Q4: List four major causes of Delayed Walking in pediatrics.

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## 3. Problem Solving (2 Cases)

**Case 1:**

A mother brings her 18-month-old male infant to the pediatric clinic because he is unable to walk alone or even stand with support. On physical examination, the infant shows a marked head lag when pulled up from a supine position, and an exaggerated "U-shape" is noted during ventral suspension.

Q1: What is your clinical definition or diagnosis for this presentation?

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Q2: Mention two lower motor unit levels or muscle disorders that can cause this presentation.

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**Case 2:**

During a routine health check-up for a 4-month-old infant, the pediatrician evaluates the developmental milestones. The infant is able to turn his head to sounds, laughs loudly, and can roll from back to front. However, on performing the neurological examination, the pediatrician notes that the Moro reflex is still fully persistent and can be easily elicited.

Q1: Based on the key development warning signs, why is this finding significant?

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Q2: List two gross motor and two social milestones that this infant should normally achieve at this exact age (4 months).

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## CHAPTER 02

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# Nutrition & Nutritional Disorders (Selected Topics)

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## 1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four anti-infective properties or factors found in human breast milk.

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Q2: Enumerate the four constant features required to diagnose Kwashiorkor (KWO).

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Q3: Enumerate the classic radiological epiphyseal triad seen in an active wrist X-ray of a child with Rickets.

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## 2. Short Answered Questions (Definitions, Mention, List)

Q4: Mention the absolute maternal and infant contraindications to breastfeeding.

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Q5: Define Marasmus according to the Welcome and anthropometric criteria.

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## 3. Problem Solving (2 Cases)

**Case 1:**

A 15-month-old female infant is brought to the clinic due to progressive failure to thrive and a change in her behavior. On examination, she is characteristically apathetic and never smiles. She has marked pitting edema in both feet and legs, and a prominent "moon face" appearance. Her subcutaneous fat seems relatively preserved, but her hair is dry, easily pickable, and shows alternating bands of normal coloration and hypopigmentation.

Q1: What is the accurate clinical diagnosis for this child?

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Q2: What is the medical term for the alternating hair hypopigmentation bands, and what does it signify?

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**Case 2:**

A 10-month-old exclusively breastfed infant who lives in an apartment with no sun exposure is evaluated for delayed motor milestones. On examination, the infant has excessive frontal sweating and is highly irritable. Skeletal examination reveals an abnormal softness over the occipital bones, prominent widening of both wrists and ankles, and a visible palpable enlargement of the costochondral junctions on the chest wall.

Q1: What is your provisional diagnosis for this infant?

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Q2: Based on your curriculum, state the medical term for the abnormal skull softness, and predict the expected serum levels of Calcium and Phosphorus in this active phase.

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## CHAPTER 03

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# G.I.T. (Selected Topics)

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## 1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four clinical diagnostic Red Flags that suggest Organic Abdominal Pain in children.

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Q2: Enumerate four acute clinical manifestations of IgE-Mediated Cow's Milk Allergy (CMA).

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## 2. Short Answered Questions (Definitions, Mention, List)

Q3: Define "The Acute Abdomen" and list its three key local physical signs.

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Q4: Differentiate accurately between the definitions of "Vomiting" and "Regurgitation".

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## 3. Problem Solving (2 Cases)

**Case 1:**

An 8-year-old girl presents to the clinic with a history of recurrent episodes of abdominal pain over the past 4 months. The pain has a gradual onset and is characteristically localized in the periumbilical region with a poor relationship to food intake or defecation. She is completely symptom-free between episodes. On physical examination, her growth is normal, and there is no localized abdominal tenderness or organomegaly.

Q1: What is the most likely diagnosis for this child?

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Q2: Mention two associated autonomic symptoms that can be seen in about 30 percent of such cases.

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**Case 2:**

A 3-month-old formula-fed male infant is brought to the outpatient clinic with chronic diarrhea, failure to thrive, and frequent streaks of blood and mucus in his stool. Physical examination reveals chronic, severe atopic dermatitis on his cheeks that is refractory to local treatment. The symptoms started 3 weeks after introducing standard cow's milk formula.

Q1: What is the clinical diagnosis for this infant?

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Q2: Mention the gold standard diagnostic test to confirm this condition, and state the first-line formula recommended for this formula-fed infant.

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**1. Short Essay & Enumerate (4 Points)**

Q1: Enumerate four indications or limitations where Oral Rehydration Solution (ORS) is considered NOT suitable or has limitations in management.

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Q2: Enumerate four clinical diagnostic "Red Flags" that a mother should look for to immediately seek medical hospital evaluation for a child with diarrhea.

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## 2. Short Answered Questions (Definitions, Mention, List)

Q3: Define "Dysentery" and differentiate between acute, persistent, and chronic diarrhea based on duration.

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Q4: List four medical (infective or non-infective) etiologies that can cause Acute Diarrhea in children.

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## 3. Problem Solving (2 Cases)

**Case 3:**

A 10-month-old infant is brought to the dehydration center with a 2-day history of acute watery diarrhea and frequent vomiting. On examination, the infant is highly restless and irritable. His eyes appear characteristically sunken. When the mother offers water, the infant drinks it thirstily and eagerly. When the pediatrician performs a skin pinch over the abdomen, the skin turgor goes back slowly.

Q1: What is the accurate classification of dehydration for this infant, and what is the appropriate management plan?

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Q2: Based on your curriculum, how should the ORS be administered, and what should be done if the infant vomits during administration?

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**Case 4:**

A 5-week-old male infant presents with nonbilious projectile vomiting immediately after feeding. The mother states that the emesis has been progressive over the last week, and the infant is always intensely hungry and wants to feed again immediately after vomiting. On physical examination, the infant shows signs of severe dehydration, visible gastric peristalsis is noted over the abdomen, and an olive-shaped hard mass is clearly palpated above and to the right of the umbilicus.

Q1: What is the clinical diagnosis, and what is the diagnostic thickness or length criteria required to confirm it on an Ultrasound examination?

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Q2: State the expected laboratory acid-base disturbance for this condition, and mention the surgical procedure of choice.

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## CHAPTER 04

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# Genetic Diseases (Selected Topics)

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## 1. Short Essay & Enumerate (4 Points)

Q1: Enumerate four dysmorphic features seen in the face and head of a child with Down Syndrome.

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Q2: Enumerate four clinical indications for performing Prenatal Diagnosis in a pregnant woman.

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## 2. Short Answered Questions (Definitions, Mention, List)

Q3: Define Turner Syndrome and mention its most common genotype.

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Q4: List four invasive or non-invasive types/methods used for Prenatal Diagnosis.

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## 3. Problem Solving (2 Cases)

**Case 1:**

A 41-year-old pregnant female gives birth to a male newborn. On physical examination, the neonate shows generalized hypotonia, a flat occiput, upward slanting palpebral fissures, a flat midface with a protruding tongue, and a single transverse palmar crease on both hands. Auscultation of the heart reveals a loud pansystolic murmur at the lower left sternal border.

Q1: What is your provisional diagnosis for this neonate?

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Q2: Mention the most common cytogenetic type responsible for this condition and state its recurrence risk.

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**Case 2:**

A 14-year-old girl is brought to the pediatric endocrine clinic due to failure to achieve menarche and severe short stature. On examination, her intelligence is completely normal, but she has a short webbed neck, a broad chest with widely spaced nipples, and a wide carrying angle at her elbows. Her blood pressure is persistently elevated at 140/90 mmHg.

Q1: What is the most likely clinical diagnosis?

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Q2: Based on your curriculum, explain the importance of prenatal screening for this disorder using the Maternal Serum Triple Test.

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